

Kunal Garg

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Ph.D. candidate in Economics with a strong quantitative and computational background, experienced in large-scale data analysis, structural and reduced-form methods, numerical optimization, and causal inference, with strong Python proficiency and exposure to machine learning techniques.

EDUCATION

Ph.D. in Economics	TEXAS A&M UNIVERSITY, 2022 -- 2027 (EXPECTED)
M.Sc. in Quantitative Economics	INDIAN STATISTICAL INSTITUTE (DELHI), 2018 -- 2020
B.A. in Economics	UNIVERSITY OF DELHI, 2015 -- 2018

PROFESSIONAL EXPERIENCE

Research Assistant	DEPARTMENT OF ECONOMICS, TEXAS A&M UNIVERSITY, 2023/06 -- 2025/08
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- Built establishment-level record linkage between a Houston payment-card merchant panel and Harris County Appraisal District property records, designing a two-tier scheme that separated a high-precision match targeting a sub-1% false-match rate from a wider-coverage nearest-neighbor match, and validating match quality by drawing random samples and adjudicating them by hand.
- Implemented a structural model of firm entry under strategic interaction in Python, solving a sequential-move entry game as a nested fixed point with under-relaxed iteration, estimating it by maximum likelihood with numerical integration over the unobserved quality distribution, and holding the simulation draws fixed across parameter evaluations so the objective stayed smooth.
- Encoded state unemployment-benefit rules into a firm-level eligibility dataset, deriving weekly and maximum benefit amounts from federal industry wage statistics under the statutory formula, its benefit floor and ceiling, and its two competing maximum calculations.

Research Associate	CAFRAL, RESERVE BANK OF INDIA, 2021/05 -- 2022/04
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- Supported research on post-crisis regulatory forbearance and zombie lending in Indian banking: re-estimated triple-difference specifications across updated data vintages and reviewed identification choices on bank-firm matched supervisory credit data (~38,000 firm-bank-year observations).
- Worked with Indian firm-level and supervisory banking data — CMIE Prowess, the RBI's CRILC credit registry, Basic Statistical Returns and the Database on Indian Economy — pulling updated vintages and re-running the panel specifications.
- Prototyped a CNN in Python to classify Sentinel-2 daylight imagery against VIIRS nightlights-derived activity bins, as an exploratory test of remote-sensing methods for measuring local economic activity where survey data is unavailable.

Research Associate	GRADUATE SCHOOL OF BUSINESS, KREA UNIVERSITY, 2021/03 -- 2021/04
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- Supported field implementation of a randomized trial of gender-sensitization training for police officers in Bihar, India.

RESEARCH EXPERIENCE

"Belief Stabilization and the Flood Insurance Gap: Evidence from the National Flood Insurance Program"	JOB MARKET PAPER, SOLE-AUTHORED, ONGOING
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- Built a longitudinal policy-level panel from FEMA's public NFIP records — 67 million policy terms across 57 states and territories, with no native policy identifier — by constructing a property-level linkage key, deduplicating within and across state partitions, and validating the split row-for-row against the source, yielding a ~53.5 million term panel spanning 2011–2019.
- Found that households' retention response to local flood claims decays with policy tenure — a 3.2 percentage point gap at first renewal falling to zero by year seven — and decomposed it into a fading neighbor-claim effect and an own-claim effect that runs the opposite way and strengthens with tenure.
- Estimated a tenure-dependent Bayesian belief-updating model of insurance renewal by simulated method of moments (coarse grid then Nelder-Mead), with bootstrapped moment covariance, Jacobian-based analytic standard errors, an overidentification test, parameter recovery on simulated data, and four policy counterfactuals.

 github.com/gargk98/credit-risk-default-prediction

- Coauthored a working paper benchmarking a dynamic logit (discrete-time hazard, Campbell et al. tradition) against XGBoost for corporate default prediction, and built the full pipeline — data construction, estimation and evaluation — on a 157,556 firm-year US panel under a strictly out-of-time design. Discrimination stays stable across regimes (AUC-ROC 0.90–0.93) while calibrated probability levels fail inside macro-shock episodes in opposite directions: understating defaults by roughly half in 2007–08, overstating by an order of magnitude in 2020–22.
- Estimated a pooled dynamic logit with standard errors clustered by firm alongside an XGBoost classifier on identical features, tuning over 50 randomized configurations with a 5-fold time-series split scored on AUC-PR, correcting for a 70:1 class imbalance, and applying Platt scaling to calibrate the frozen booster.
- Translated model output into expected-loss terms using Basel Foundation IRB loss-given-default and debt-proxied exposure, and set the calibration findings against supervisory model-risk expectations (SR 26-2), CCAR/DFAST stress testing and IFRS 9 / CECL provisioning practice.

“Corporate Board Management and Greenhouse Gas Emissions” WORKING PAPER WITH COAUTHORS, ONGOING

- Examines whether mandatory greenhouse gas disclosure changes firm emissions and board composition, and whether effects spread to unregulated firms through shared directors.
- Built the analysis dataset, resolving entities across EPA facility-level pollution registries (Toxic Release Inventory, Greenhouse Gas Reporting Program), BoardEx governance records and Compustat — four sources with no shared identifier — on cleaned company names across a 127,000-firm universe.
- Showed that synthetic control fit quality across a firm-level panel was mechanically best precisely where the outcome variable sat near zero, establishing that a favorable pre-treatment fit statistic can be an artifact of a degenerate outcome rather than evidence of a valid comparison group.

TEACHING EXPERIENCE

Instructor, “The Organization of Industry” TEXAS A&M UNIVERSITY, FALL 2025

- Designed and taught an undergraduate course covering market structure, pricing, and strategic behavior, as sole instructor.

Teaching Assistant and Grader TEXAS A&M UNIVERSITY, FALL 2022, SPRING 2023, SPRING 2026

- Ran office hours and graded for undergraduate microeconomic theory and economic development.
- Graded four substantial Python projects and several in-class assignments for an applied data science in economics course, reading student code directly and writing the grading rubric from the assignment briefs and reference scripts.

Teaching Fellow DEPARTMENT OF ECONOMICS, ASHOKA UNIVERSITY, 2020/08 – 2021/05

- Supported four undergraduate and graduate economics courses: ran weekly discussion sections across two sections each of statistics and mathematics for economists, wrote and graded the statistics exams, and graded for development economics and a graduate Python programming elective.

AWARDS, GRANTS AND PRESENTATIONS

- Lechner Graduate Grant, awarded by the Department of Economics, Texas A&M University, 2022–23.
- Presented the job market paper at the Southern Economic Association Annual Meeting, Tampa, November 2025, supported by a departmental travel grant.

SKILLS

Programming and tools: Python (pandas, NumPy, SciPy, scikit-learn, XGBoost), Stata, R, MATLAB, SQL (familiar), L^AT_EX, Git, HPC/SLURM, QGIS.

Methods and data: causal inference, structural estimation, survival and hazard modeling, record linkage at scale, model calibration and out-of-time validation; Compustat, CRSP, WRDS.

Languages: English (fluent), Hindi (native).